LAB 4

NAME: Hajra Sarwar
REG NO: 2023-BSE-022
CLASS: BSE-5A
SUBJECT: CC

LAB TITLE: Virtualization & Linux Fundamentals

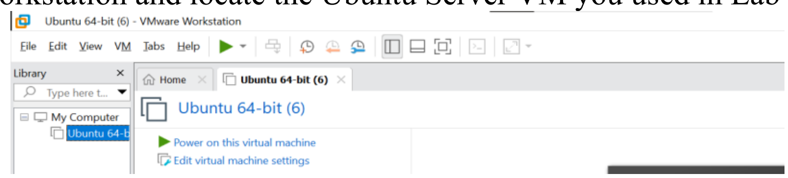
TASK:

Task 1: Verify VM resources in VMware

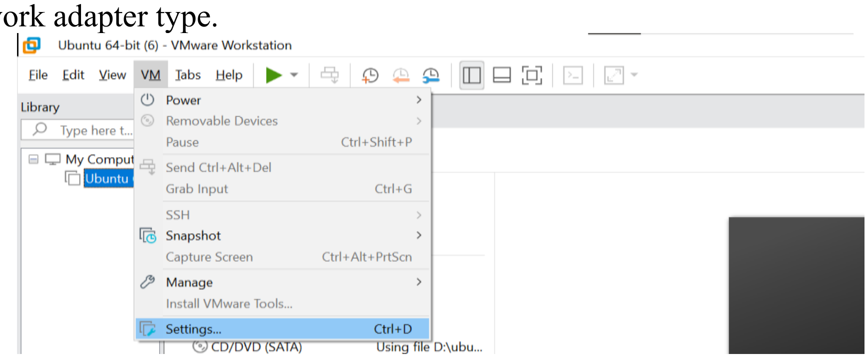
Confirm the VM resources that were allocated in Lab 1.

Steps

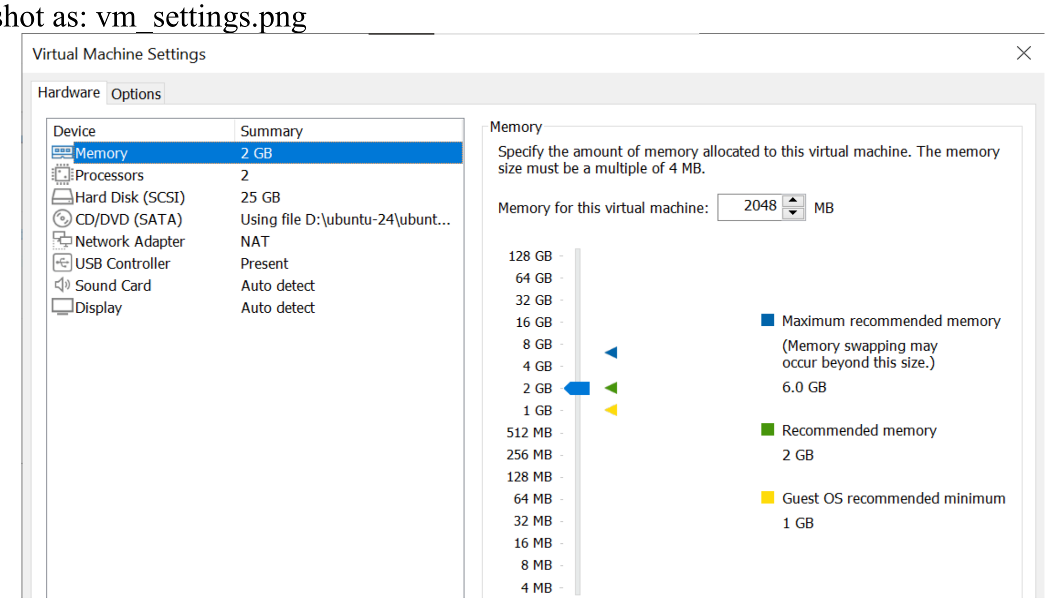
1. Open VMware Workstation and locate the Ubuntu Server VM you used in Lab 1.



2. Inspect VM settings and note the following (no commands required for GUI): VM name, RAM, CPU, disk, and network adapter type.



3. Take a screenshot of the VM settings window showing RAM, CPU, disk and networking. Save screenshot as: vm_settings.png

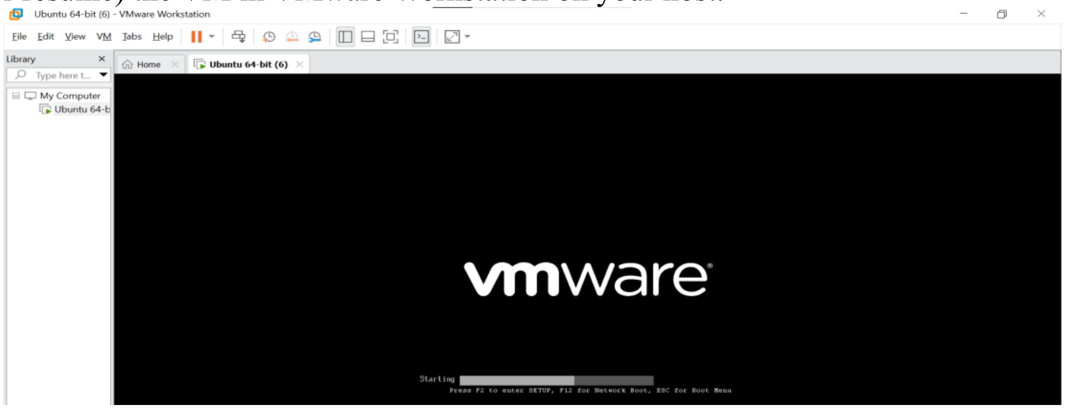


Task 2: Start VM and log in (use your preferred host terminal method only)

Use a single preferred host-terminal method to connect to the VM. Do not switch between methods during the task.

Steps

1. Start (or resume) the VM in VMware Workstation on your host.



2. From your host, open your preferred terminal (for example: Windows Command Prompt, PowerShell, macOS Terminal, or Linux Terminal) and connect to the VM using SSH. Example:
   ssh student@<vm-ip-address>

   1. Find the IP address of your Ubuntu Server using "ip addr"

```
ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 00:0c:29:6a:f6:87 brd ff:ff:ff:ff:ff:ff
  altname enp2s1
  inet 192.168.161.129/24 metric 100 brd 192.168.161.255 scope global dynamic ens33
     valid_lft 1082sec preferred_lft 1082sec
  inet6 fe80::20c:29ff:fe6a:f687/64 scope link
     valid_lft forever preferred_lft forever
```

   2. Connect via SSH from Windows

```
PS C:\Users\HP> ping 192.168.161.129

Pinging 192.168.161.129 with 32 bytes of data:
Reply from 192.168.161.129: bytes=32 time=1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time<1ms TTL=64
Reply from 192.168.161.129: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.161.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms
```

   3. **Accept the fingerprint** (first time only)
      Type yes when prompted.

```
PS C:\Users\HP> ssh hajra@192.168.161.129
The authenticity of host '192.168.161.129 (192.168.161.129)' can't be established.
ED25519 key fingerprint is SHA256:M5unwK7+jxUhr6KM3bbMPRkghlGd+Kq5uspTQZ5Up3w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

   4. **Enter your password**
      Use the same password you set up during the Ubuntu Server installation.

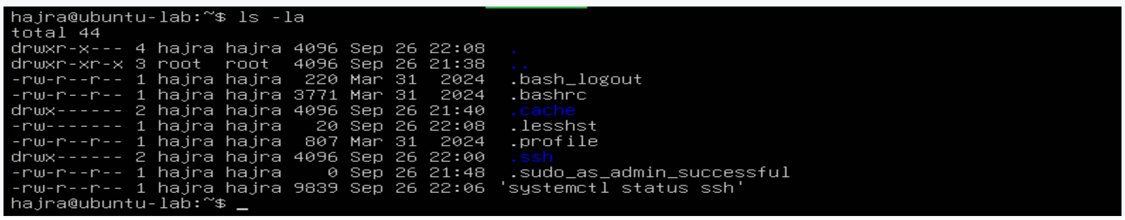
```
Warning: Permanently added '192.168.161.129' (ED25519) to the list of known hosts.
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-84-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat 27 Sep 10:28:45 UTC 2025

  System load:  1.46               Processes:             276
  Usage of /:   70.2% of 11.21GB   Users logged in:       1
  Memory usage: 57%                IPv4 address for ens33: 192.168.161.129
  Swap usage:   9%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


hajra@ubuntu-lab:~$ _
```

3. After logging in, run both commands and capture them together in a single screenshot:
whoami
pwd

Task 3: Filesystem exploration — root tree and dotfiles

Steps (run inside VM terminal)

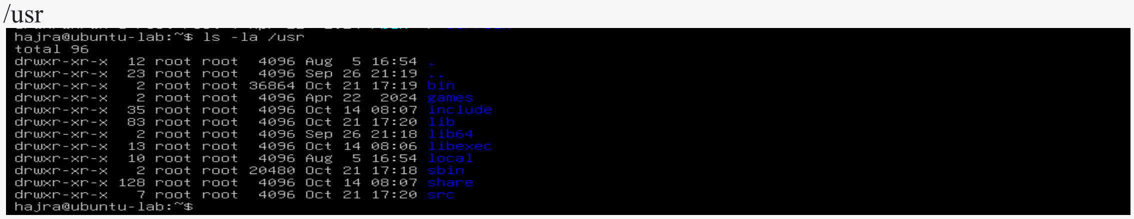
1. List root directory contents:
ls -la /



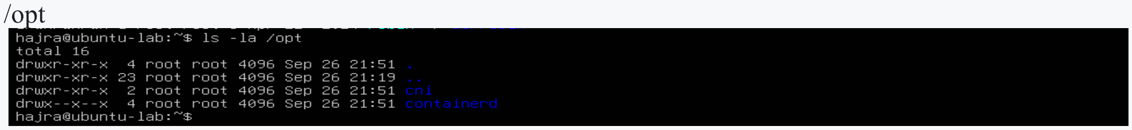
2. Inspect these directories (run each command and screenshot the output):
ls -la /bin

ls -la /sbin

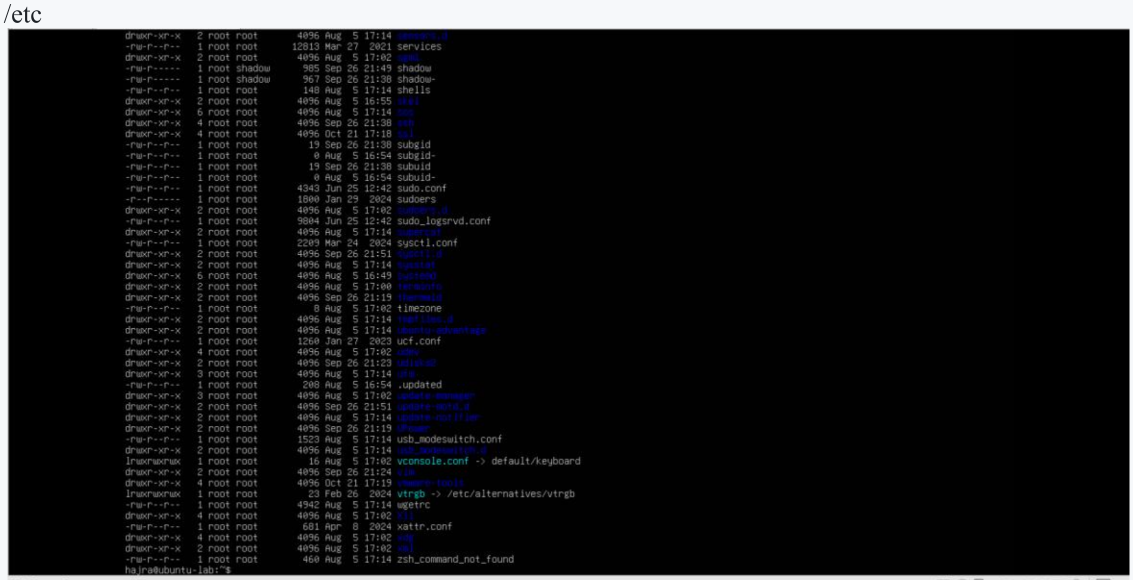
ls -la /usr



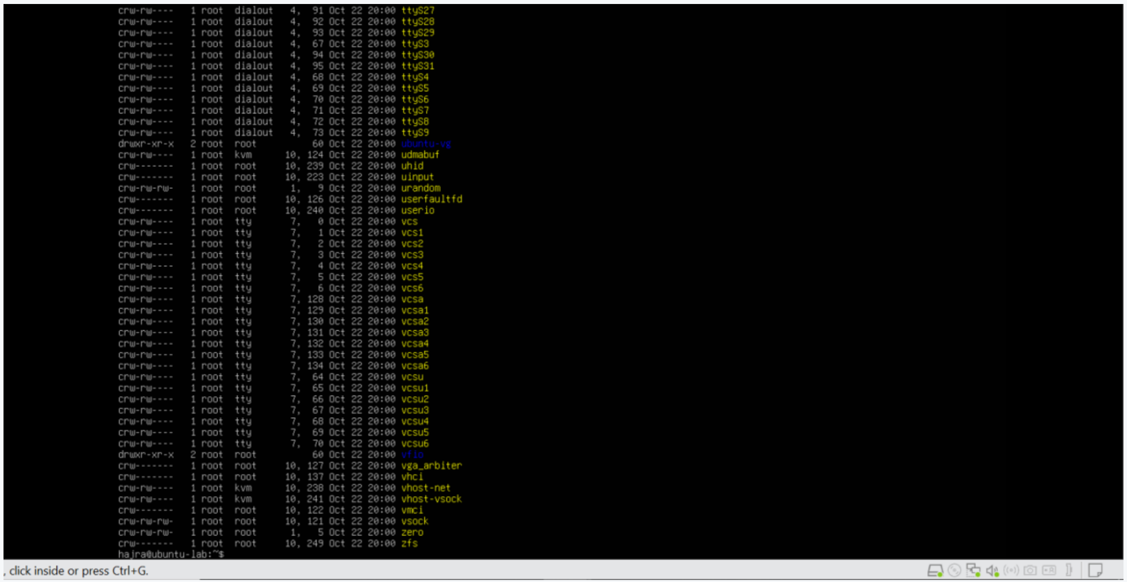
ls -la /opt



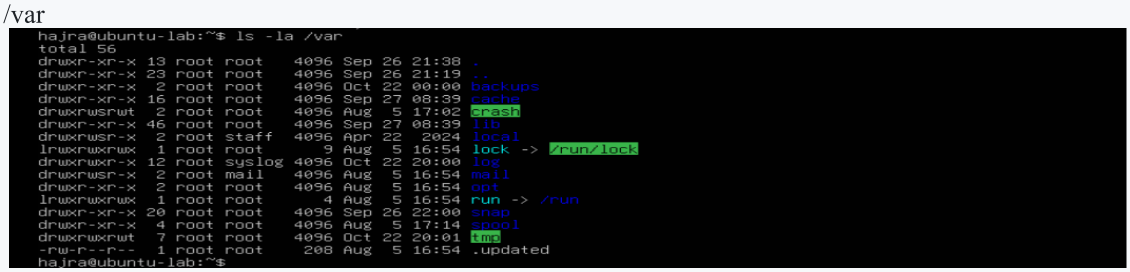
ls -la /etc



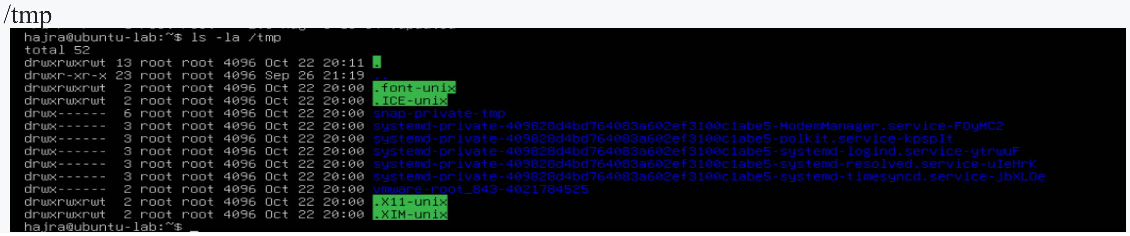
ls -la /dev



ls -la /var

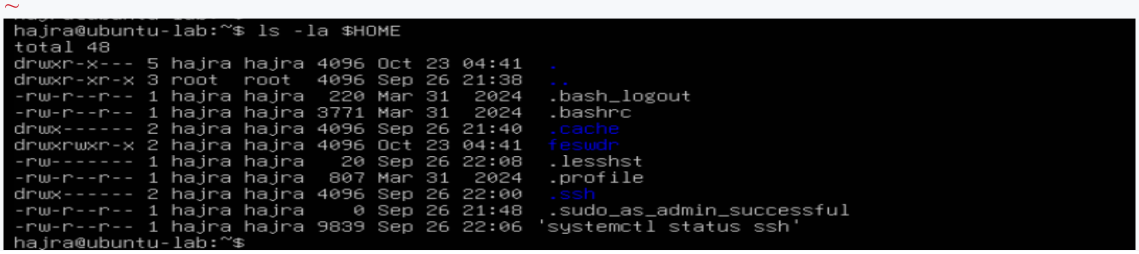


ls -la /tmp



3. List your home directory and show hidden (dot) files:

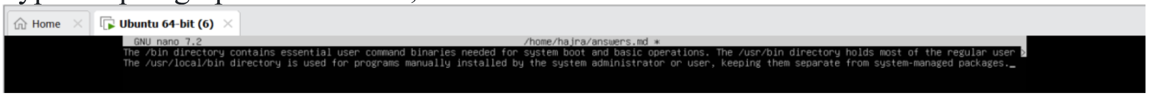
ls -la ~



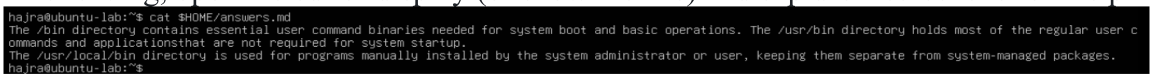
4. Write a short paragraph (3–5 sentences) that explains the difference between /bin, /usr/bin and /usr/local/bin. Open your editor:

nano ~/answers.md

- Type the paragraph in the editor, save and exit.



- After saving, open the editor display (or show the file) and capture a screenshot of the paragraph.



Task 4: Essential CLI Tasks — Navigation and File Operations

Steps Performed

1. **Create a Workspace and Navigate**

mkdir -p ~/lab4/workspace/python_project

Created a new workspace directory for the Python project.

cd ~/lab4/workspace/python_project

Navigated into the newly created directory.

Pwd

Verified the current working directory path.

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2. **Create Files Using an Editor**

nano README.md

Added the text **"Lab 4 README"** and saved the file.

nano main.py

Added the Python code: print("hello lab4") and saved the file.

nano .env

Added the line **ENV=lab4** and saved the file.

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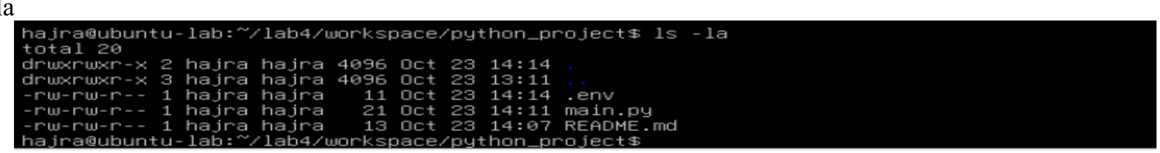
3. **List Files**

ls -la



Displayed all files, including hidden ones, in the current directory.

---

4. **Copy, Move, and Remove Files**

cp README.md README.copy.md

Created a copy of the README file.

mv README.copy.md README.dev.md

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ mv README.md.copy.md README.dev.md
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Renamed (moved) the copied file.

rm README.dev.md

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ rm README.dev.md
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Deleted the renamed file.

---

5. **Work with Directories**

mkdir -p ~/lab4/workspace/java_app

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ mkdir -p $HOME/laab4/workspace/java_app
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

Created another directory for a Java app.

cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ cp -r $HOME/lab4/workspace/python_project $HOME/lab4/workspace/java_app_copy
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Copied the entire Python project directory recursively.

ls -la ~/lab4/workspace

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ ls -la $HOME/lab4/workspace
total 16
drwxrwxr-x 4 hajra hajra 4096 Oct 23 14:34 .
drwxrwxr-x 3 hajra hajra 4096 Oct 23 13:11 ..
drwxrwxr-x 2 hajra hajra 4096 Oct 23 14:34 java_app_copy
drwxrwxr-x 2 hajra hajra 4096 Oct 23 14:27 python_project
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

Verified the copied directories.

---

6. **Use Command History and Tab Completion**

History

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ history
    1  ls -la $HOME
    2  nano $HOME/answers.md
    3  cat $HOME/answers.md
    4  mkdir -p $HOME/lab4/workspace/python_project
    5  cd $HOME/lab4/workspace/python_project
    6  pwd
    7  nano README.md
    8  nano main.py
    9  nano .env
   10  ls -la
   11  cp README.md README.md.copy.md
   12  mv README.copy.md README.dev.md
   13  mv README.md.copy.md README.dev.md
   14  rm README.dev.md
   15  mkdir -p $HOME/laab4/workspace/java_app
   16  cp -r $HOME/lab4/workspace/python_project $HOME/lab4/workspace/java_app_copy
   17  ls -la $HOME/lab4/workspace
   18  history
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Displayed a list of previously executed commands.

- Demonstrated tab completion by typing part of a file or directory name and pressing **Tab** to auto-complete it.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ cat main.py
print ('hello lab4')
```

<u>Task 5:</u> System info, resources & processes

Collect system information and observe processes. Use screenshots only.

Steps (inside VM terminal)

1. Kernel and OS:
   uname -a

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ uname -a
Linux ubuntu-lab 6.8.0-85-generic #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 18 15:26:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

2. CPU (ensure model name visible):
   cat /proc/cpuinfo

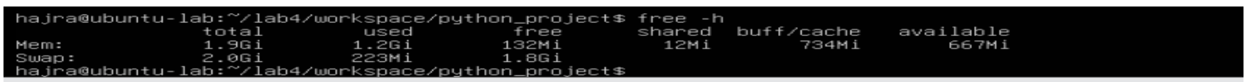
```
Home    Ubuntu 64-bit (6)
core id         : 0
cpu cores       : 1
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 22
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc
 arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_time
r aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap clflus
hopt xsaveopt xsavec xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
bugs            : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit srbds mmio_stale_data retbleed gds bhi
bogomips        : 3791.99
clflush size    : 64
cache_alignment : 64
address sizes   : 45 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id       : GenuineIntel
cpu family      : 6
model           : 142
model name      : Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz
stepping        : 10
microcode       : 0xffffffff
cpu MHz         : 1895.997
cache size      : 6144 KB
physical id     : 2
siblings        : 1
core id         : 0
cpu cores       : 1
apicid          : 2
initial apicid  : 2
fpu             : yes
fpu_exception   : yes
cpuid level     : 22
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_tsc
 arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_time
r aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap clflus
hopt xsaveopt xsavec xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
bugs            : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit srbds mmio_stale_data retbleed gds bhi
bogomips        : 3791.99
clflush size    : 64
cache_alignment : 64
address sizes   : 45 bits physical, 48 bits virtual
power management:

hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

1, click inside or press Ctrl+G.

3. Memory:
   free -h



```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       1.2Gi       132Mi        12Mi       734Mi       667Mi
Swap:          2.0Gi       223Mi       1.8Gi
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

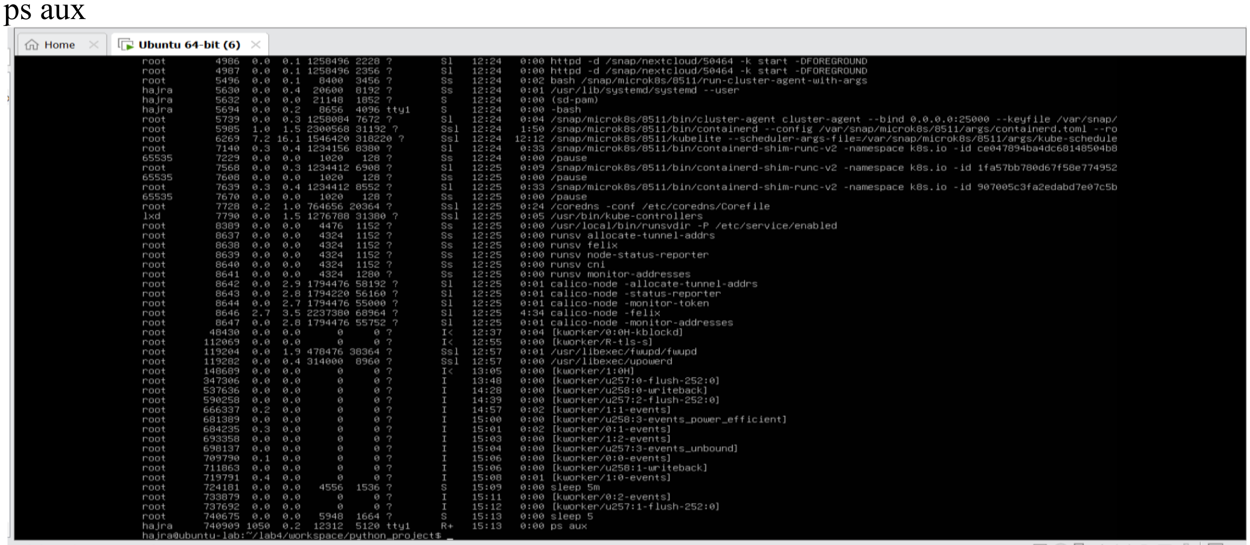
4. Disk:
   df -h

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       1.2Gi       132Mi        12Mi       734Mi       667Mi
Swap:          2.0Gi       223Mi       1.8Gi
hajra@ubuntu-lab:~/lab4/workspace/python_project$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              192M  1.6M  191M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   12G   11G  343M  97% /
tmpfs                              960M     0  960M   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  192M  1.6G  11% /boot
tmpfs                              192M   20K  192M   1% /run/user/1000
shm                                 64M     0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/ce047894ba4dc68148504b884d
dbe81bfeef7c1524cd31f3f71c0866c562fc57/shm
shm                                 64M     0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/907005c3fa2edabd7e07c5bb55
f734ead13372884c9de10e7c3258af5f4114e6/shm
shm                                 64M     0   64M   0% /var/snap/microk8s/common/run/containerd/io.containerd.grpc.v1.cri/sandboxes/1fa57bb780d67f58e774952f1e
e1858625b771458a25591d41300447ba4e2367/shm
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

5. View OS release information:
   cat /etc/os-release

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.3 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.3 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

6. Processes (show top lines of ps output):
   ps aux



```
Home    Ubuntu 64-bit (6)
root        4986  0.0  0.1 1258496 2228 ?      Sl   12:24   0:00 httpd -d /snap/nextcloud/50464 -k start -DFOREGROUND
root        4987  0.0  0.1 1258496 2356 ?      Sl   12:24   0:00 httpd -d /snap/nextcloud/50464 -k start -DFOREGROUND
root        5496  0.0  0.1   8400  3456 ?      Ss   12:24   0:02 bash /snap/microk8s/8511/run-cluster-agent-with-args
hajra       5630  0.0  0.4  20600  8192 ?      Ss   12:24   0:01 /usr/lib/systemd/systemd --user
hajra       5632  0.0  0.0  21148  1852 ?      S    12:24   0:00 (sd-pam)
hajra       5694  0.0  0.2   8656  4096 tty1   S    12:24   0:00 -bash
root        5739  0.0  0.3 1258084 7672 ?      Sl   12:24   0:04 /snap/microk8s/8511/bin/cluster-agent cluster-agent --bind 0.0.0.0:25000 --keyfile /var/snap/
root        5985  1.0  1.5 2300568 31192 ?     Ssl  12:24   1:50 /snap/microk8s/8511/bin/containerd --config /var/snap/microk8s/8511/args/containerd.toml --ro
root        6269  7.2 16.1 1546420 318220 ?    Ssl  12:24  12:12 /snap/microk8s/8511/kubelite --scheduler-args-file=/var/snap/microk8s/8511/args/kube-schedule
root        7140  0.3  0.4 1234156 8380 ?      Sl   12:24   0:33 /snap/microk8s/8511/bin/containerd-shim-runc-v2 -namespace k8s.io -id ce047894ba4dc68148504b8
65535       7229  0.0  0.0   1020   128 ?      Ss   12:24   0:00 /pause
root        7568  0.0  0.3 1234412 6908 ?      Sl   12:25   0:09 /snap/microk8s/8511/bin/containerd-shim-runc-v2 -namespace k8s.io -id 1fa57bb780d67f58e774952
65535       7608  0.0  0.0   1020   128 ?      Ss   12:25   0:00 /pause
root        7639  0.3  0.4 1234412 8552 ?      Sl   12:25   0:33 /snap/microk8s/8511/bin/containerd-shim-runc-v2 -namespace k8s.io -id 907005c3fa2edabd7e07c5b
65535       7670  0.0  0.0   1020   128 ?      Ss   12:25   0:00 /pause
root        7728  0.2  1.0 764656 20364 ?      Ssl  12:25   0:24 /coredns -conf /etc/coredns/Corefile
lxd         7790  0.0  1.5 1276788 31380 ?     Ssl  12:25   0:05 /usr/bin/kube-controllers
root        8389  0.0  0.0   4476  1152 ?      Ss   12:25   0:00 /usr/local/bin/runsvdir -P /etc/service/enabled
root        8637  0.0  0.0   4324  1152 ?      Ss   12:25   0:00 runsv allocate-tunnel-addrs
root        8638  0.0  0.0   4324  1152 ?      Ss   12:25   0:00 runsv felix
root        8639  0.0  0.0   4324  1152 ?      Ss   12:25   0:00 runsv node-status-reporter
root        8640  0.0  0.0   4324  1152 ?      Ss   12:25   0:00 runsv cni
root        8641  0.0  0.0   4324  1280 ?      Ss   12:25   0:00 runsv monitor-addresses
root        8642  0.0  2.9 1794476 58192 ?     Sl   12:25   0:01 calico-node -allocate-tunnel-addrs
root        8643  0.0  2.8 1794220 56160 ?     Sl   12:25   0:01 calico-node -status-reporter
root        8644  0.0  2.7 1794476 55000 ?     Sl   12:25   0:01 calico-node -monitor-token
root        8646  2.7  3.5 2237380 68964 ?     Sl   12:25   4:34 calico-node -felix
root        8647  0.0  2.8 1794476 55752 ?     Sl   12:25   0:01 calico-node -monitor-addresses
root       48430  0.0  0.0      0     0 ?      I<   12:37   0:04 [kworker/0:0H-kblockd]
root      112069  0.0  0.0      0     0 ?      I<   12:55   0:00 [kworker/R-tls-s]
root      119204  0.0  1.9 478476 38364 ?      Ssl  12:57   0:01 /usr/libexec/fwupd/fwupd
root      119282  0.0  0.4 314000  8960 ?      Ssl  12:57   0:00 /usr/libexec/upowerd
root      148689  0.0  0.0      0     0 ?      I<   13:05   0:00 [kworker/1:0H]
root      347306  0.0  0.0      0     0 ?      I    13:48   0:00 [kworker/u257:0-flush-252:0]
root      537636  0.0  0.0      0     0 ?      I    14:28   0:00 [kworker/u258:0-writeback]
root      590258  0.0  0.0      0     0 ?      I    14:39   0:00 [kworker/u257:2-flush-252:0]
root      666337  0.2  0.0      0     0 ?      I    14:57   0:02 [kworker/1:1-events]
root      681389  0.0  0.0      0     0 ?      I    15:00   0:00 [kworker/u258:3-events_power_efficient]
root      684235  0.3  0.0      0     0 ?      I    15:01   0:02 [kworker/0:1-events]
root      693358  0.0  0.0      0     0 ?      I    15:03   0:00 [kworker/1:2-events]
root      698137  0.0  0.0      0     0 ?      I    15:04   0:00 [kworker/u257:3-events_unbound]
root      709790  0.1  0.0      0     0 ?      I    15:06   0:00 [kworker/0:0-events]
root      711863  0.0  0.0      0     0 ?      I    15:06   0:00 [kworker/u258:1-writeback]
root      719791  0.4  0.0      0     0 ?      I    15:08   0:01 [kworker/1:0-events]
root      724181  0.0  0.0   4556  1536 ?      S    15:09   0:00 sleep 5m
root      733879  0.0  0.0      0     0 ?      I    15:11   0:00 [kworker/0:2-events]
root      737692  0.0  0.0      0     0 ?      I    15:12   0:00 [kworker/u257:1-flush-252:0]
root      740675  0.0  0.0   5948  1664 ?      S    15:13   0:00 sleep 5
hajra     740909 1050  0.2  12312  5120 tty1   R+   15:13   0:00 ps aux
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

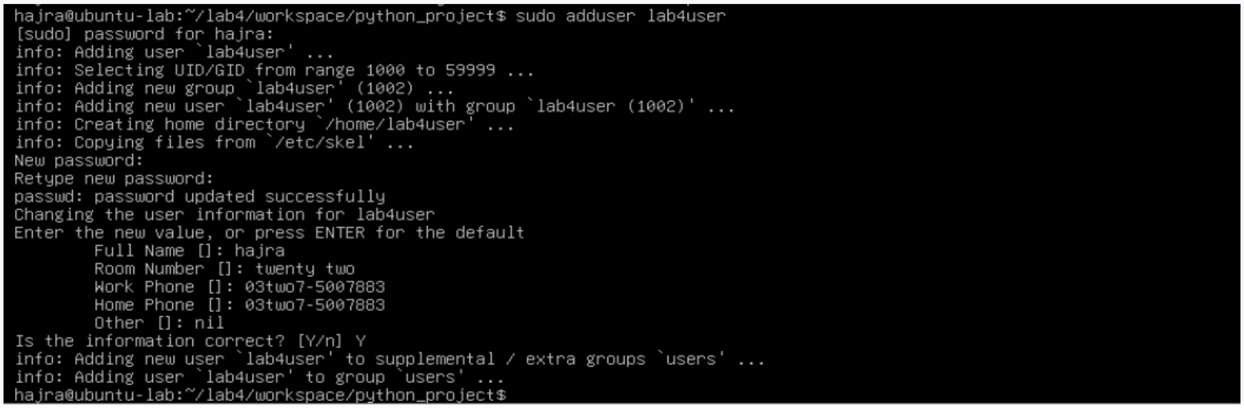
M, click inside or press Ctrl+G.

<u>Task 6:</u> Users and account verification (no sudo group change)

Steps (inside VM terminal)

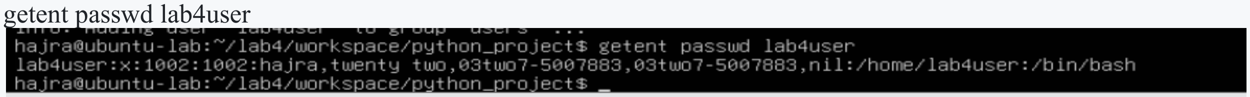
1. Create a new user named lab4user:

sudo adduser lab4user



2. Verify the user entry:

getent passwd lab4user

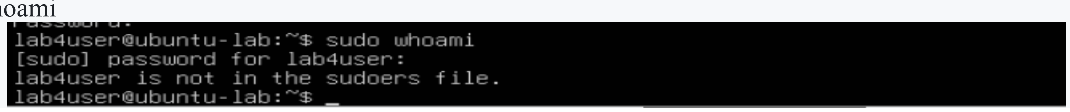


3. Switch to the new user to verify login:

su - lab4user

4. From the new user you may attempt a sudo command to show that sudo is not available for this account (expected failure), e.g.:

sudo whoami

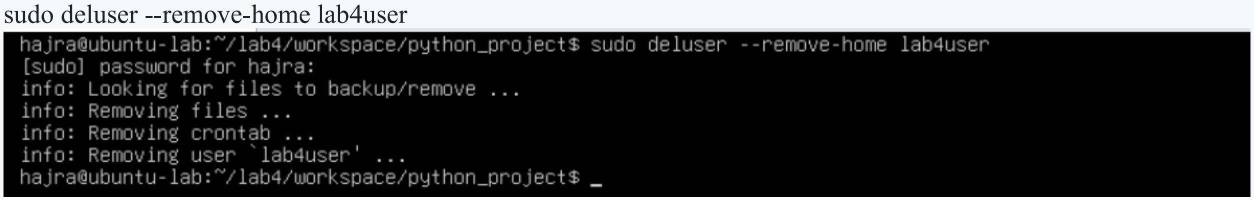


5. Return to the original user:

Exit

6. (Optional) Remove the test user when finished:

sudo deluser --remove-home lab4user

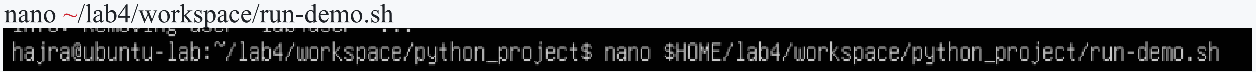


<u>Bonus Task 7</u>: Create a small demo script using an editor and run it

Steps (inside VM)

1. Open an editor to create the script:

nano ~/lab4/workspace/run-demo.sh



- Type the following lines into the editor (manually or paste), save and exit:

```
#!/bin/bash
echo "Lab 4 demo: current user is $(whoami)"
echo "Current time: $(date)"
uptime
```

free -h

```
  GNU nano 7.2                                   /home/hajra/lab4/workspace/python_project/run-demo.sh *
#!/bin/bash
echo 'Lab 4 demo: current user is $(whoami)'
echo 'Currrent time: $(date)'
uptime
free -h_
```

2. Make the script executable:

chmod +x ~/lab4/workspace/run-demo.sh

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ chmod +x ~/lab4/workspace/python_project/run-demo.sh
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

3. Run the script as your regular user:

~/lab4/workspace/run-demo.sh

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ ~/lab4/workspace/python_project/run-demo.sh
Lab 4 demo: current user is $(whoami)
Currrent time: $(date)
 19:02:29 up  6:39,  1 user,  load average: 0.78, 0.74, 0.76
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       1.2Gi       111Mi       740Ki       730Mi       652Mi
Swap:          2.0Gi       225Mi       1.8Gi
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

4. Optionally run it with sudo:

sudo ~/lab4/workspace/run-demo.sh

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ sudo ~/lab4/workspace/python_project/run-demo.sh
[sudo] password for hajra:
Lab 4 demo: current user is $(whoami)
Currrent time: $(date)
 19:03:47 up  6:41,  1 user,  load average: 0.57, 0.66, 0.73
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       1.2Gi       128Mi       824Ki       731Mi       671Mi
Swap:          2.0Gi       225Mi       1.8Gi
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Exam Evaluation Questions

1. Remote Access Verification (Cyber Login Check)

Scenario:

You are part of a SOC (Security Operations Center) investigating unauthorized access to a Linux server hosted on VMware. Prove you can securely connect and verify your identity.

Steps:

1. Connect to the Ubuntu VM remotely from your host terminal.

```
Warning: Permanently added '192.168.161.129' (ED25519) to the list of known hosts.
hajra@192.168.161.129's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-84-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat 27 Sep 10:28:45 UTC 2025

  System load:  1.46               Processes:              276
  Usage of /:   70.2% of 11.21GB   Users logged in:        1
  Memory usage: 57%                IPv4 address for ens33: 192.168.161.129
  Swap usage:   9%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


hajra@ubuntu-lab:~$ _
```

2. Verify your current user and home directory path.

```
hajra@ubuntu-lab:~$ whoami
hajra
hajra@ubuntu-lab:~$ pwd
/home/hajra
hajra@ubuntu-lab:~$
```

3. Confirm you are connected to the correct host machine.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ hostname
ubuntu-lab
```

---

2. Filesystem Inspection for Forensic Evidence

Scenario:

The incident response team suspects malicious files in system directories. You must explore the filesystem to locate and document the system's structure.

Steps:

1. Display the contents of the root directory.

```
hajra@ubuntu-lab:~$ ls -la
total 44
drwxr-x--- 4 hajra hajra 4096 Sep 26 22:08 .
drwxr-xr-x 3 root  root  4096 Sep 26 21:38 ..
-rw-r--r-- 1 hajra hajra  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 hajra hajra 3771 Mar 31  2024 .bashrc
drwx------ 2 hajra hajra 4096 Sep 26 21:40 .cache
-rw------- 1 hajra hajra   20 Sep 26 22:08 .lesshst
-rw-r--r-- 1 hajra hajra  807 Mar 31  2024 .profile
drwx------ 2 hajra hajra 4096 Sep 26 22:00 .ssh
-rw-r--r-- 1 hajra hajra    0 Sep 26 21:48 .sudo_as_admin_successful
-rw-r--r-- 1 hajra hajra 9839 Sep 26 22:06 'systemctl status ssh'
hajra@ubuntu-lab:~$ _
```

2. Display the OS version and release information.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ uname -a
Linux ubuntu-lab 6.8.0-85-generic #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 18 15:26:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

3. Explore and record directory listings for /bin, /sbin, /usr, /opt, /etc, /dev, /var, and /tmp.

```
Home   Ubuntu 64-bit (6)
lrwxrwxrwx   1 root root     7 Apr 22  2024 /bin -> usr/bin
lrwxrwxrwx   1 root root     8 Apr 22  2024 /sbin -> usr/sbin

/dev:
total 4
drwxr-xr-x  20 root  root       4500 Oct 23 12:23 .
drwxr-xr-x  23 root  root       4096 Sep 26 21:19 ..
crw-r--r--   1 root  root    10, 235 Oct 23 12:23 autofs
drwxr-xr-x   2 root  root        700 Oct 23 12:23 block
drwxr-xr-x   2 root  root         80 Oct 23 12:22 bsg
crw-rw----   1 root  disk    10, 234 Oct 23 12:23 btrfs-control
drwxr-xr-x   3 root  root         60 Oct 23 12:22 bus
lrwxrwxrwx   1 root  root          3 Oct 23 12:23 cdrom -> sr0
drwxr-xr-x   2 root  root       3740 Oct 23 12:27 char
crw--w----   1 root  tty      5,   1 Oct 23 12:23 console
lrwxrwxrwx   1 root  root         11 Oct 23 12:22 core -> /proc/kcore
drwxr-xr-x   4 root  root         80 Oct 23 12:23 cpu
crw-------   1 root  root    10, 123 Oct 23 12:23 cpu_dma_latency
crw-------   1 root  root    10, 203 Oct 23 12:23 cuse
drwxr-xr-x  10 root  root        200 Oct 23 12:23 disk
brw-rw----   1 root  disk   252,   0 Oct 23 12:23 dm-0
drwxr-xr-x   2 root  root         60 Oct 23 12:22 dma_heap
crw-rw----+  1 root  audio   14,   9 Oct 23 12:23 dmmidi
drwxr-xr-x   3 root  root        100 Oct 23 12:23 dri
crw-------   1 root  root    10, 125 Oct 23 12:23 ecryptfs
crw-rw----   1 root  video   29,   0 Oct 23 12:23 fb0
lrwxrwxrwx   1 root  root         13 Oct 23 12:22 fd -> /proc/self/fd
crw-rw-rw-   1 root  root     1,   7 Oct 23 12:23 full
crw-rw-rw-   1 root  root    10, 229 Oct 23 12:23 fuse
crw-------   1 root  root   241,   0 Oct 23 12:23 hidraw0
crw-------   1 root  root    10, 228 Oct 23 12:23 hpet
drwxr-xr-x   2 root  root          0 Oct 23 12:23 hugepages
crw-------   1 root  root    10, 183 Oct 23 12:23 hwrng
lrwxrwxrwx   1 root  root         12 Oct 23 12:23 initctl -> /run/initctl
drwxr-xr-x   4 root  root        260 Oct 23 12:23 input
crw-r--r--   1 root  root     1,  11 Oct 23 12:23 kmsg
lrwxrwxrwx   1 root  root         28 Oct 23 12:23 log -> /run/systemd/journal/dev-log
brw-rw----   1 root  disk     7,   0 Oct 23 12:23 loop0
brw-rw----   1 root  disk     7,   1 Oct 23 12:23 loop1
brw-rw----   1 root  disk     7,  10 Oct 23 12:23 loop10
brw-rw----   1 root  disk     7,  11 Oct 23 12:23 loop11
brw-rw----   1 root  disk     7,  12 Oct 23 12:23 loop12
brw-rw----   1 root  disk     7,  13 Oct 23 12:23 loop13
brw-rw----   1 root  disk     7,  14 Oct 23 12:23 loop14
brw-rw----   1 root  disk     7,  15 Oct 23 12:23 loop15
brw-rw----   1 root  disk     7,  16 Oct 23 12:23 loop16
brw-rw----   1 root  disk     7,  17 Oct 23 12:23 loop17
brw-rw----   1 root  disk     7,  18 Oct 23 12:23 loop18
brw-rw----   1 root  disk     7,  19 Oct 23 12:23 loop19
:_
, click inside or press Ctrl+G.
```

4. Display all hidden files in your home directory.

```
hajra@ubuntu-lab:~$ ls -la $HOME
total 48
drwxr-x--- 5 hajra hajra 4096 Oct 23 04:41 .
drwxr-xr-x 3 root  root  4096 Sep 26 21:38 ..
-rw-r--r-- 1 hajra hajra  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 hajra hajra 3771 Mar 31  2024 .bashrc
drwx------ 2 hajra hajra 4096 Sep 26 21:40 .cache
drwxrwxr-x 2 hajra hajra 4096 Oct 23 04:41 feswdr
-rw------- 1 hajra hajra   20 Sep 26 22:08 .lesshst
-rw-r--r-- 1 hajra hajra  807 Mar 31  2024 .profile
drwx------ 2 hajra hajra 4096 Sep 26 22:00 .ssh
-rw-r--r-- 1 hajra hajra    0 Sep 26 21:48 .sudo_as_admin_successful
-rw-r--r-- 1 hajra hajra 9839 Sep 26 21:06 'systemctl status ssh'
hajra@ubuntu-lab:~$
```

5. Create a markdown file summarizing your findings on key binary directories.
    1. Go to your **home directory** (or your lab workspace folder):
       cd ~ or cd ~/lab4/workspace

       ```
       hajra@ubuntu-lab:~$ cd ~/lab4/workspace
       ```

    2. Create and open a new Markdown file using **nano editor**:
       nano report.md

       ```
       hajra@ubuntu-lab:~/lab4/workspace$ nano report.md_
       ```

    3. Inside nano, type your short summary.

       ```
       GNU nano 7.2                                   report.md *
       <!--System Directory Summary-->
       - **/bin** - contains essential user command binaries (like ls, cp, mv)
       - **/sbin** - Contain system binaries used mainly by the root user for system admi
       - **/usr** - Contains user-installed programs, libraries, and documentation.
       - **/opt** - Used for optional or third-party software.
       - **etc** - Contains configuration files for the system and installed services.
       - **/dev** - Contains device files representing hardware devices.
       - **/var** - Contains variable data such as logs, caches, and spool files.
       - **tmp** - Temporary files created by running processes.
       ```

    4. When done, **save and exit nano**.
    5. Verify your file exists and display its contents:
       cat report.md

```
hajra@ubuntu-lab:~/lab4/workspace$ cat report.md
<!--System Directory Summery-->
- **/bin** - contains essential user command binaries (like ls, cp, mv)
- **/sbin** - Contain system binaries used mainly by the root user for system admi
- **/usr** - Contains user-installed programs, libraries, and documentation.
- **/opt** - Used for optional or third-party software.
- **etc** - Contains configuration files for the system and installed services.
- **/dev** - Contains device files representing hardware devices.
- **/var** - Contains variable data such as logs, caches, and spool files.
- **tmp** - Temporary files created by running processes.

hajra@ubuntu-lab:~/lab4/workspace$ _
```

---

3. Evidence Handling & File Operations

Scenario:

You are creating a sandbox environment to safely analyze and handle suspicious files collected from a compromised system.

Steps:

1. Create a structured folder hierarchy under your home directory for analysis.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ pwd
/home/hajra/lab4/workspace/python_project
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

2. Create three text files, including one hidden file, in your workspace.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ ls -la
total 20
drwxrwxr-x 2 hajra hajra 4096 Oct 23 14:14 .
drwxrwxr-x 3 hajra hajra 4096 Oct 23 13:11 ..
-rw-rw-r-- 1 hajra hajra   11 Oct 23 14:14 .env
-rw-rw-r-- 1 hajra hajra   21 Oct 23 14:11 main.py
-rw-rw-r-- 1 hajra hajra   13 Oct 23 14:07 README.md
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

3. Create a backup copy of one file, rename it, and then delete it after verification.

cp README.md README.copy.md

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ cp README.md README.md.copy.md
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

Created a copy of the README file.

mv README.copy.md README.dev.md

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ mv README.md.copy.md README.dev.md
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Renamed (moved) the copied file.

rm README.dev.md

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ rm README.dev.md
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Deleted the renamed file.

4. Copy the entire workspace as an evidence backup folder.

mkdir -p ~/lab4/workspace/java_app

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ mkdir -p $HOME/laab4/workspace/java_app
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

Created another directory for a Java app.

cp -r ~/lab4/workspace/python_project ~/lab4/workspace/java_app_copy

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ cp -r $HOME/lab4/workspace/python_project $HOME/lab4/workspace/java_app_copy
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

Copied the entire Python project directory recursively.

ls -la ~/lab4/workspace

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ ls -la $HOME/lab4/workspace
total 16
drwxrwxr-x 4 hajra hajra 4096 Oct 23 14:34 .
drwxrwxr-x 3 hajra hajra 4096 Oct 23 13:11 ..
drwxrwxr-x 2 hajra hajra 4096 Oct 23 14:34 java_app_copy
drwxrwxr-x 2 hajra hajra 4096 Oct 23 14:27 python_project
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

Verified the copied directories.

5. Display your command history to document all actions performed.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ history
    1  ls -la $HOME
    2  nano $HOME/answers.md
    3  cat $HOME/answers.md
    4  mkdir -p $HOME/lab4/workspace/python_project
    5  cd $HOME/lab4/workspace/python_project
    6  pwd
    7  nano README.md
    8  nano main.py
    9  nano .env
   10  ls -la
   11  cp README.md README.md.copy.md
   12  mv README.copy.md README.dev.md
   13  mv README.md.copy.md README.dev.md
   14  rm README.dev.md
   15  mkdir -p $HOME/laab4/workspace/java_app
   16  cp -r $HOME/lab4/workspace/python_project $HOME/lab4/workspace/java_app_copy
   17  ls -la $HOME/lab4/workspace
   18  history
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

6. Demonstrate Linux auto-completion by typing a partial command or filename.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ cat main.py
print ('hello lab4')
```

---

4. System Profiling and Process Monitoring

Scenario:

You are investigating a potential malware infection that is consuming excessive resources on the Linux VM.

Steps:

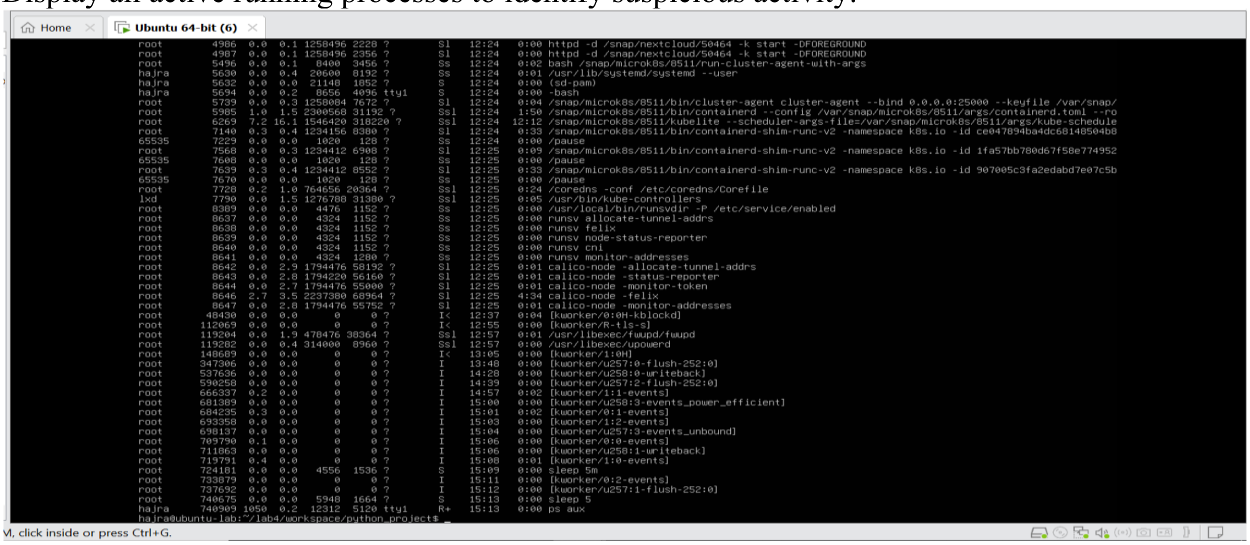
1. Display the system's OS and kernel version for the investigation report.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ uname -a
Linux ubuntu-lab 6.8.0-85-generic #85-Ubuntu SMP PREEMPT_DYNAMIC Thu Sep 18 15:26:59 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

2. Display CPU, memory, and disk usage information.

```
hajra@ubuntu-lab:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           1.9Gi       938Mi       322Mi       9.9Mi       830Mi       981Mi
Swap:          2.0Gi        30Mi       2.0Gi
hajra@ubuntu-lab:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
tmpfs                              192M  1.5M  191M   1% /run
/dev/mapper/ubuntu--vg-ubuntu--lv   12G   11G  335M  97% /
tmpfs                              960M     0  960M   0% /dev/shm
tmpfs                              5.0M     0  5.0M   0% /run/lock
/dev/sda2                          2.0G  192M  1.6G  11% /boot
tmpfs                              192M   20K  192M   1% /run/user/1000
hajra@ubuntu-lab:~$ _
```

3. Display all active running processes to identify suspicious activity.



5. User Account Audit & Privilege Escalation Simulation

Scenario:

You are performing a user activity audit on a compromised Linux server. The SOC suspects a newly created account (lab4user) may have been used for unauthorized access. Your task is to simulate the account creation, perform privilege tests, and analyze authentication logs for forensic evidence.

Steps:

1. Create a new test user named lab4user.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ sudo adduser lab4user
[sudo] password for hajra:
info: Adding user `lab4user' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `lab4user' (1002) ...
info: Adding new user `lab4user' (1002) with group `lab4user (1002)' ...
info: Creating home directory `/home/lab4user' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for lab4user
Enter the new value, or press ENTER for the default
        Full Name []: hajra
        Room Number []: twenty two
        Work Phone []: 03two7-5007883
        Home Phone []: 03two7-5007883
        Other []: nil
Is the information correct? [Y/n] Y
info: Adding new user `lab4user' to supplemental / extra groups `users' ...
info: Adding user `lab4user' to group `users' ...
hajra@ubuntu-lab:~/lab4/workspace/python_project$
```

2. Verify that the new user record exists in the system's user database.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ getent passwd lab4user
lab4user:x:1002:1002:hajra,twenty two,03two7-5007883,03two7-5007883,nil:/home/lab4user:/bin/bash
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

3. Log in as lab4user and confirm successful login.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ su - lab4user
Password:
lab4user@ubuntu-lab:~$ _
```

4. Attempt to run an administrative command as lab4user (expect permission denied).

```
lab4user@ubuntu-lab:~$ sudo whoami
[sudo] password for lab4user:
lab4user is not in the sudoers file.
lab4user@ubuntu-lab:~$ _
```

5. Switch back to your main analyst account.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```

6. (Optional) Remove the lab4user account after the audit and verify deletion.

```
hajra@ubuntu-lab:~/lab4/workspace/python_project$ sudo deluser --remove-home lab4user
[sudo] password for hajra:
info: Looking for files to backup/remove ...
info: Removing files ...
info: Removing crontab ...
info: Removing user `lab4user' ...
hajra@ubuntu-lab:~/lab4/workspace/python_project$ _
```